

A speed limit for cheap monitoring in AI control

An experimental test of when correlated monitor scores break the safety case

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1. The problem

In AI control (Greenblatt et al. 2023), a cheap *trusted monitor* scores every action of an untrusted model and a scarce human audit adjudicates the few it flags; the binding resource is human attention, so the audit budget is a false-alarm rate α and the figure of merit is the catch rate at that budget. Safety analyses estimate this rate from the monitor’s score *histograms* — the single-step separation $d = \text{KL}(\text{attack} \parallel \text{benign})$ of the one-step score distributions — reasoning as if successive actions were independent. But one agent executing one plan produces correlated scores that drift, persist, and cluster: aggregating suspicion across a multi-step attack roughly doubles the catch rate (Bhatt et al. 2025, Ctrl-Z), because a real attack must look suspicious *repeatedly*. A safety case that ignores how scores accumulate can be not merely loose but unsafe.

2. What the theory establishes

The governing quantity is not the single-step d but the relative-entropy *rate* $R = \lim_n \frac{1}{n} \text{KL}(P_1^{(n)} \parallel P_0^{(n)})$, where $P_1^{(n)}$ and $P_0^{(n)}$ are the n -step laws of the attack-active and benign score streams; $R = d$ for independent scores, and in general the sign of $d - R$ is model-dependent. Casting trusted monitoring as adversarial quickest change-detection, a complete manuscript (drafted) proves:

- **Exact speed limit.** Every monitor whose mean time between false alarms is at least $1/\alpha$ has optimal worst-case detection delay $(1 \pm o(1)) \log(1/\alpha)/R$ — a lower bound, attained by a CUSUM (cumulative-sum) detector. Equivalently, an attack of length L is caught with probability $\rightarrow 1$ if $LR > \log(1/\alpha)$, and missed by *every* budget-compliant monitor if $LR < \log(1/\alpha)$. The rate R governs, not the histogram separation d .
- **Soundness.** The histogram (“independence”) safety case is sound *iff* $d \leq R$. When $d > R$ it certifies uncatchable attacks as safe: for a Gaussian AR(1) score model with lag-1 autocorrelation $\phi \in (-1, 1)$, $R = d(1 - \phi)/(1 + \phi)$, so the margin in attack length is $d/R = (1 + \phi)/(1 - \phi) - 9\times$ at $\phi = 0.8$, unbounded as $\phi \rightarrow 1$. When $d < R$ — e.g. a correlation-only attack with benign-identical marginals, $d = 0 < R$ — it calls a catchable attack invisible.

The sign of $d - R$, unseen by histograms, is the one bit a safety case must establish.

The machinery is classical (Page 1954; Lorden 1971; Lai 1998); the contribution is the control framing, the rate R as the safety-relevant quantity, and the d -vs- R characterisation — now settled, which makes the experiment pointed.

3. The experimental programme — the core of the Fellowship work

The theory makes sharp, falsifiable predictions about *real* monitors, and the groundwork is in place. A synthetic pipeline (AR(1), two-state, and renewal simulators) validates estimators for d , R , the autocorrelation function (ACF), and the catch-rate frontier against closed-form ground truth; and the real-trajectory path is wired end-to-end — BashArena through ControlArena/Inspect, whose logs expose a *per-action* suspicion score with attack-span labels, extracted per step and fed to the d , R , ACF, and bootstrap estimators; a model-backed monitor is plumbed in, so only a real LLM-monitor run remains in place of the current deterministic scaffold. This settles the feasibility crux — real pipelines expose the per-action stream the theory needs — and the study then runs as:

- **Autocorrelation first, the leading indicator.** The ACF of per-step scores — cheap, prior to any rate estimation — reveals whether temporal dependence, hence $R \neq d$, exists in real traces at all: a flat ACF means $R \approx d$ (the independence case is sound here); slow decay is where $R < d$ lives.
- **The sign of $d - R$, the empirical crux.** Estimate d from histograms and R from a model-free rate estimator, on benign and attack trajectories long enough that LR can straddle $\log(1/\alpha)$. This one bit decides whether the standard independence safety case is unsound in practice.
- **The catch-rate frontier.** Empirical catch-rate-vs-budget against the independence prediction e^{-Ld} and the corrected e^{-LR} ; predicted to track the latter, a factor $\approx e^{L(d-R)}$ below what the histogram analysis promises.
- **A designed attack, as measurement.** That correlated multi-step attacks evade monitors is already known; the test here is *quantitative* — a red-team policy that tunes score autocorrelation and pacing should degrade detection by the predicted factor d/R , tracing out the closed-form curve rather than merely confirming the effect.
- **Conservative, falsifiable certification.** For a usable safety case, a high-confidence *lower* bound on R (over-estimating R is the unsafe direction; the theory supplies a model-light one). The study is decision-relevant either way — $d \approx R$ or $d < R$ in real monitors would show the independence case already safe.

Fit and risk.

The project pairs a settled theory with a concrete measurement on real control transcripts, matching mentors in monitoring, control, and scalable oversight (Roger; Benton; Khan). Rate estimation is data-hungry — met by the conservative lower bound above and the synthetic harness with closed-form ground truth that calibrates the estimators before real data; open theory (adaptive timing; unknown-attack-class detection) is future work, not a blocker.